

A Unified Physics-Based Compact Model and Calibrated Circuit-Validation Flow for Multi-Material CFET Logic: Si, SiGe, MoS₂/WSe₂, CNT and GaN

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Abstract—Complementary FET (CFET) stacking removes the n/p lateral separation from the standard-cell footprint, but the choice of channel material for the two tiers is still open. We present a unified, physics-based compact model (UCM-CFET) whose single equation core (EKV-type smooth overdrive, velocity-saturation-limited drift-diffusion current with a tanh drain-voltage interpolation, charge-conserving terminal charges, series resistance and temperature dependence) is extended by material blocks: quantum capacitance and trap-limited ideality for two-dimensional MoS₂/WSe₂ channels, a Landauer quasi-ballistic branch with a metallic-tube shunt for carbon-nanotube (CNT) arrays, and an electro-thermal node for GaN. Nine device models (Si GAA n/p, SiGe p, MoS₂ n, WSe₂ p, CNT n/p, GaN n and projected p-GaN) are written once as Verilog-A for Cadence Spectre/ADE, mirrored exactly (within 0.1%) by an open-source ngspice behavioural library, and calibrated by least squares to literature-anchored reference I - V data with 1.3–12.1% mean absolute percentage error on I_D - V_G and 3.7–12.6% on I_D - V_D families. The calibrated models drive a common CFET inverter cell with explicit inter-tier coupling capacitance, output parasitic and local-interconnect resistance, 5/7/9-stage ring oscillators (ROs), unified V_{DD} and temperature sweeps, material-specific sensitivity studies and Monte Carlo yield analysis. At nominal supply the 5-stage ROs run at 19.39, 21.39, 2.59, 45.39 and 1.69 GHz for the Si, SiGe, TMD, CNT and GaN platforms with 2.09, 2.12, 0.67, 0.84 and 94.17 fJ per cycle. A contact-resistance crossover map shows the TMD CFET keeps an iso-frequency energy advantage over Si only below a critical contact resistance; the CNT platform is limited by metallic-tube leakage above 10^{-3} purity defects; the GaN platform shows a frequency temperature coefficient of nan ppm/K between 300 and 500 K. The outcome is a material-selection map rather than a single winner, together with conceptual, DRC-clean GDSII layouts of the five inverter and RO cells.

Index Terms—CFET, compact model, Verilog-A, nanosheet, SiGe, MoS₂, WSe₂, carbon nanotube, GaN, ring oscillator, Monte Carlo, design-technology co-optimization.

I. INTRODUCTION

THE complementary FET (CFET), in which the p-channel device is stacked vertically over the n-channel device around a common gate, is the leading candidate to continue track-height scaling below the lateral gate-all-around (GAA)

nanosheet node [1], [2], [3]. Because the two tiers are fabricated sequentially or share a common stack, the CFET also opens the possibility of placing *different channel materials* in the two tiers: strained SiGe for the hole channel [4], [5], two-dimensional (2D) transition-metal dichalcogenides (TMDs) for ultimate thickness scaling [6], [7], [8], [9], [10], aligned carbon-nanotube (CNT) arrays for quasi-ballistic transport [11], [12], [13], and wide-band-gap GaN for high-temperature or power-control logic [14], [15], [16], [17].

Assessing these options at the circuit level requires compact models that are (i) physically interpretable, so that a material parameter such as contact resistance or metallic-tube fraction maps to one model parameter, (ii) numerically smooth, so that Spectre or ngspice converge in ring-oscillator (RO) transients, (iii) charge-conserving, so that delay and energy are meaningful, and (iv) calibrated by a documented procedure. Industry-standard models such as BSIM-CMG [18] satisfy (ii)–(iii) for Si but do not expose 2D, CNT or GaN physics; published 2D [19] and CNT [20] models are material-specific and are rarely compared on one footing.

This paper makes four contributions. First, a unified compact model (UCM-CFET) with one core and four material blocks, written in Verilog-A for Cadence Virtuoso/Spectre/ADE [21], [22], [23] and mirrored by an ngspice behavioural library [24] that reproduces the Verilog-A equations to within 0.1%. Second, a reproducible calibration flow with explicit targets and error metrics. Third, a complete circuit-validation flow (inverter cell with CFET parasitics, VTC, transient delay and energy, 5/7/9-stage ROs, V_{DD} and temperature sweeps, material-specific sensitivity studies, Monte Carlo yield), executed identically for all five platforms. Fourth, a material-selection map derived from those results. The platform maturity is stated explicitly (Table I): the Si and SiGe models are advanced-node, roadmap-relevant; the TMD and CNT models are emerging post-Si projections; the GaN model, in particular its p-channel device, is exploratory.

II. UNIFIED COMPACT MODEL

A. Core equations

All nine devices share the four-terminal interface (d, g, s, b; b is a reference terminal tied to the source in every circuit) and internal nodes d_i , s_i (series resistance) and t_j (temperature

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Simulation sources, Verilog-A models, Cadence scripts, GDSII layouts and result files are available at the companion repository (branch cfet_multimaterial).

TABLE I
THE FIVE CFET MATERIAL PLATFORMS AND THEIR MATURITY.

ID	Bottom (n)	Top (p)	Purpose
CFET-Si	Si GAA NS	Si GAA NS	baseline logic (advanced node series, $C_{ox,eff} = C_{ox}C_q/(C_{ox} + C_q)$, a trap-limited ideality
CFET-SiGe	Si GAA NS	SiGe GAA NS	p-FET mobility boost (advanced node)
CFET-TMD	MoS ₂	WSe ₂	low- V_{DD} , contact-limited (emerging)
CFET-CNT	n-CNTFET	p-CNTFET	quasi-ballistic post-Si (emerging)
CFET-GaN	n-GaN	p-GaN (proj.)	high- T /power-control (exploratory)

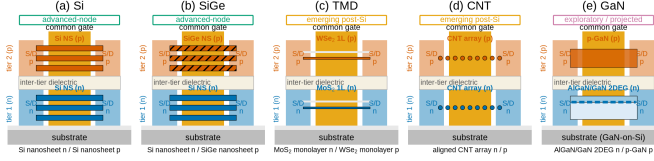


Fig. 1. Cross-section cartoons of the five CFET platforms: bottom n-tier, inter-tier dielectric, top p-tier, common gate wrapping both tiers, and n/p source/drain contacts. Maturity tags follow Table I.

rise). With $V_T = k_B T_{dev}/q$, the model uses an EKV-type smooth overdrive [25]

$$V_{ov} = 2nV_T \ln \left[1 + \exp \left(\frac{V_{GS} - V_{TH,eff}}{2nV_T} \right) \right], \quad (1)$$

so that $V_{ov}^2 \propto \exp[(V_{GS} - V_{TH})/nV_T]$ in weak inversion (subthreshold slope $nV_T \ln 10$) and $V_{ov} \rightarrow V_{GS} - V_{TH}$ in strong inversion. Drain-induced barrier lowering enters as $V_{TH,eff} = V_{TH,T} - \eta \sqrt{V_{DS}^2 + \delta^2}$ and the temperature dependence as $V_{TH,T} = V_{TH0} + k_{VTH}(T_{dev} - T_{nom})$, $\mu_T = \mu_0(T_{dev}/T_{nom})^{-m_\mu}$. The drift-diffusion branch with velocity saturation is

$$I_D = \frac{\beta V_{ov}^2}{2(1 + V_{ov}/V_{sat,v})} \tanh \left(\frac{2V_{DS}}{V_{dsat}} \right) (1 + \lambda |V_{DS}|), \quad (2)$$

with $\beta = \mu_T C_{ox,eff} W/L$, $V_{sat,v} = v_{sat} L/\mu_T$ and $V_{dsat} = V_{ov}/(1 + V_{ov}/V_{sat,v})$. The factor 2 in the tanh argument makes the low- V_{DS} conductance exactly βV_{ov} while the saturation current retains the square law for long channels and the velocity-saturated limit $C_{ox} W v_{sat} V_{ov}/2$ for short channels [26]; the function is odd in V_{DS} and infinitely differentiable, which is what the Newton iterations of a circuit simulator need. Series resistances $R_S = R_{SW}/W$ and $R_D = R_{DW}/W$ act between the external and internal nodes.

B. Terminal charges

A DC-only model cannot produce RO frequencies. Terminal charges are defined per branch,

$$Q_{gs} = \frac{1}{2} W L C_{ox,eff} V_{ov}(V_{GS}) + C_{gso} W V_{GS}, \quad (3)$$

$$Q_{gd} = \frac{1}{2} W L C_{ox,eff} V_{ov}(V_{GD}) + C_{gdo} W V_{GD}, \quad (4)$$

with V_{ov} evaluated without DIBL so that each charge depends only on its own branch voltage, and injected as $I = dQ/dt$. With $Q_g = Q_{gs} + Q_{gd}$, $Q_s = -Q_{gs}$ and $Q_d = -Q_{gd}$, charge conservation $Q_g + Q_d + Q_s = 0$ holds by construction; the symmetric source/drain-end partition reduces to 50/50 in the linear region and decouples the drain in saturation.

C. Material blocks

Si and SiGe nanosheets use the core with $W_{eff} = N_{NS} \cdot 2(W_{NS} + T_{ps})/W$. *MoS₂/WSe₂* add the quantum capacitance in series, $C_{ox,eff} = C_{ox}C_q/(C_{ox} + C_q)$, a trap-limited ideality $\eta = \eta_0/(1 + qD_{is}/C_{ox})$ and contact-dominated series resistance $R_{SW} = R_{DW} = R_C$ [19], [8]. *CNT arrays* replace (2) by a Landauer quasi-ballistic branch [20], [26]

$$I_D = N_{tube} G_0 T n V_T \left[F(\eta_s) - F\left(\eta_s - \frac{V_{DS}}{nV_T}\right) \right] (1 + \lambda |V_{DS}|) + G_{met} V_{DS}, \quad (5)$$

with $F(x) = \ln(1 + e^x)$, $\eta_s = \alpha(V_{GS} - V_{TH,eff})/nV_T$, $G_0 = 4q^2/h$, $N_{tube} = D_{CNT} W (1 - F_{met})$ and the metallic-tube shunt $G_{met} = F_{met} D_{CNT} W G_0 T$. *GaN* adds an electro-thermal node: $I(t_j) = -I_D V_{DS} + V(t_j)/R_{th} + C_{th} dV(t_j)/dt$ and $T_{dev} = T_{amb} + V(t_j)$, i.e. the quasi-static self-heating $T_{dev} = T_{amb} + R_{th} P_{diss}$ with a thermal time constant $R_{th} C_{th}$.

D. Implementation and equivalence

The nine Verilog-A modules (gaa_n, gaa_p, sig_e_p, mos2_n, wse2_p, cnt_n, cnt_p, gan_n, gan_p) are generated from one parameter table, as is an ngspice behavioural library in which the same equations are evaluated by B-sources, .func macros and charge-based ($Q =$) capacitors; intermediate quantities (V_{ov} , V_{dsat} , η_s) are solved on auxiliary nodes, which speeds the transient analysis by $7\times$ without changing the result. A NumPy reference implementation is used for calibration. DC sweeps of all nine devices agree between ngspice and the reference implementation to within 0.1% (the residual is the tolerance of the bisection used for the series-resistance loop). Spectre was not available in the computing environment used for this paper; the Cadence SKILL/OCEAN/ADE scripts that build the library, symbols, schematics and analyses are provided and documented step by step, and all manuscript results were generated with the ngspice mirror.

III. CALIBRATION

No measured or TCAD data were available, so the calibration targets are literature-anchored synthetic reference curves: for each device the on-current, off-current, subthreshold slope and DIBL at the platform V_{DD} (Table II, sources in the companion doc/calibration_targets.md) are imposed on an *independent* EKV-type formulation with a fourth-order power-law drain-voltage smoothing and explicit series resistance; the UCM is then fitted to those curves. This makes the fit a real test of the model's functional form (the two formulations differ in their saturation behaviour and charge interpolation), while keeping the targets traceable to published ranges [27], [28], [4], [5], [8], [9], [10], [11], [17], [14]. The flow accepts measured or TCAD CSV files with the same column layout as drop-in replacements.

Seven parameters are fitted per device by bounded non-linear least squares on $\log_{10} I_D$ of the linear and saturation I_D - V_G curves and on the relative error of the I_D - V_D family: V_{TH0} , μ_0 (transmission T and gate efficiency α for CNT), v_{sat} , $R_{SW} = R_{DW}$ (Si, SiGe, GaN only), λ , η and n_0 . The 2D and CNT contact resistances are held at their physical nominal

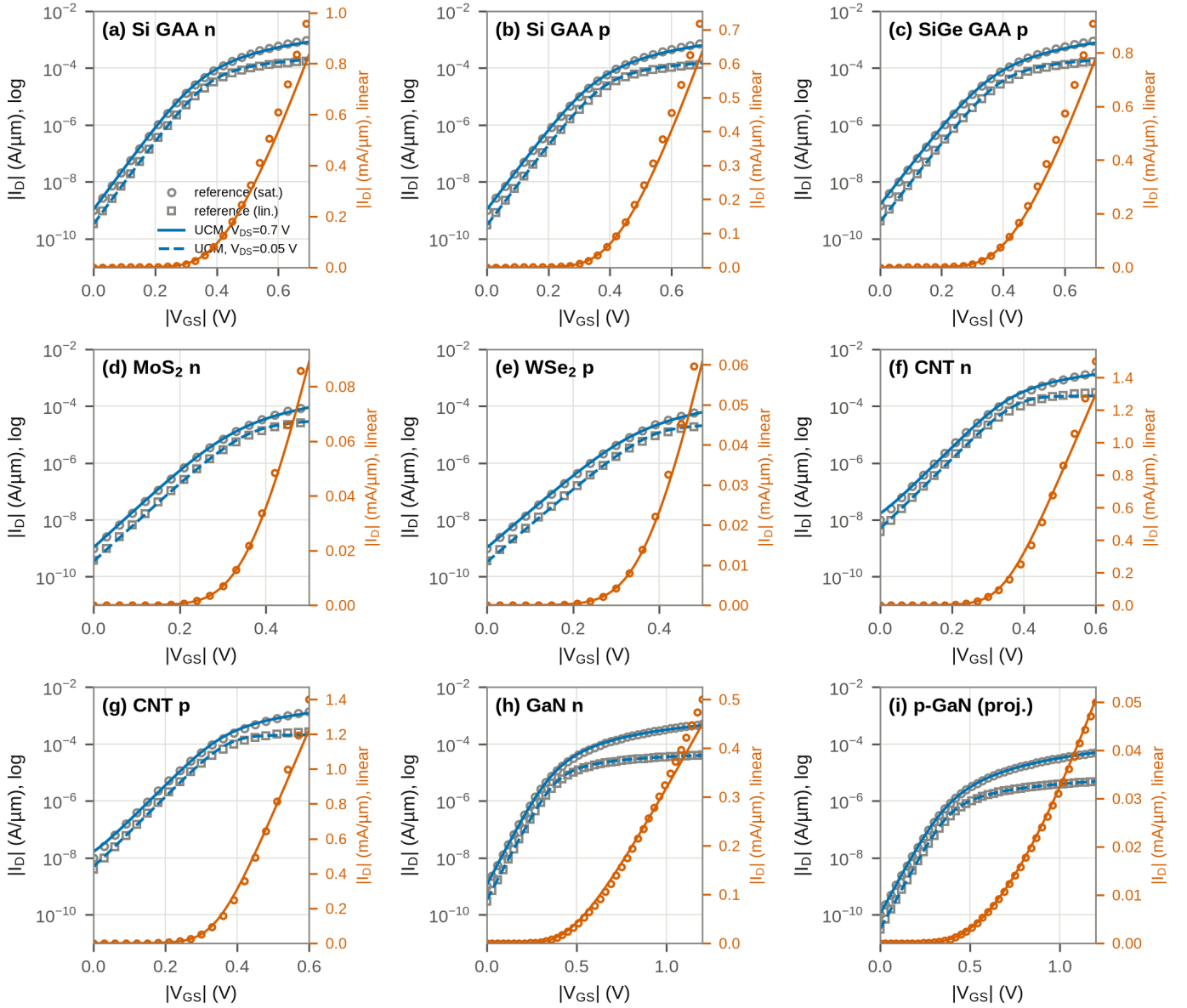


Fig. 2. I_D - V_G calibration of the nine device models (lines) against the literature-anchored reference curves (symbols) at $|V_{DS}| = 0.05$ V and $|V_{DS}| = V_{DD}$, on logarithmic (left axes) and linear (right axes) scales.

values because they are measured quantities in practice and are the subject of the sensitivity study. Figs. 2–4 and Table III summarise the result: the mean absolute percentage error (MAPE) is 1.3–12.1% on I_D - V_G and 3.7–12.6% on the I_D - V_D families, with the on/off currents within 20% and the subthreshold slopes within 3 mV/dec of their targets. The largest residuals are in the SiGe/Si output conductance (the tanh interpolation saturates harder than the reference, which the fit compensates with λ at its 0.3 V^{-1} bound) and in the CNT subthreshold region, where the single-ideality Landauer branch cannot reproduce the reference's 75 mV/dec slope together with its DIBL.

IV. CIRCUIT-VALIDATION FLOW

A. Testbenches

The electrical inverter (Fig. 5a) is a conventional CMOS inverter; its CFET interpretation is carried by the parasitic network: C_{stack} between the n- and p-tier drain nodes, C_{out} from the output to V_{SS} and $R_{local,n/p}$ in the two drain legs. These are design variables (0.05/0.3 fF/10 Ω for Si and SiGe, 0.03/0.3 fF/50 Ω for TMD and CNT, 0.1/1 fF/20 Ω for GaN, each the mid-value of the ranges in Section II of the flow definition). The same five inverter cells are used for the VTC (DC sweep of V_{in} from 0 to V_{DD} : switching voltage V_M , maximum gain A_V , noise margins from the unity-gain points, static supply current), the transient delay (pulse $0 \rightarrow V_{DD}$, 100 ps delay, 5 ps edges, 500 ps width, 1 ns period, $C_{load} = 0.3$ fF or 1 fF for GaN; t_{pHL} , t_{pLH} at the 50% crossings;

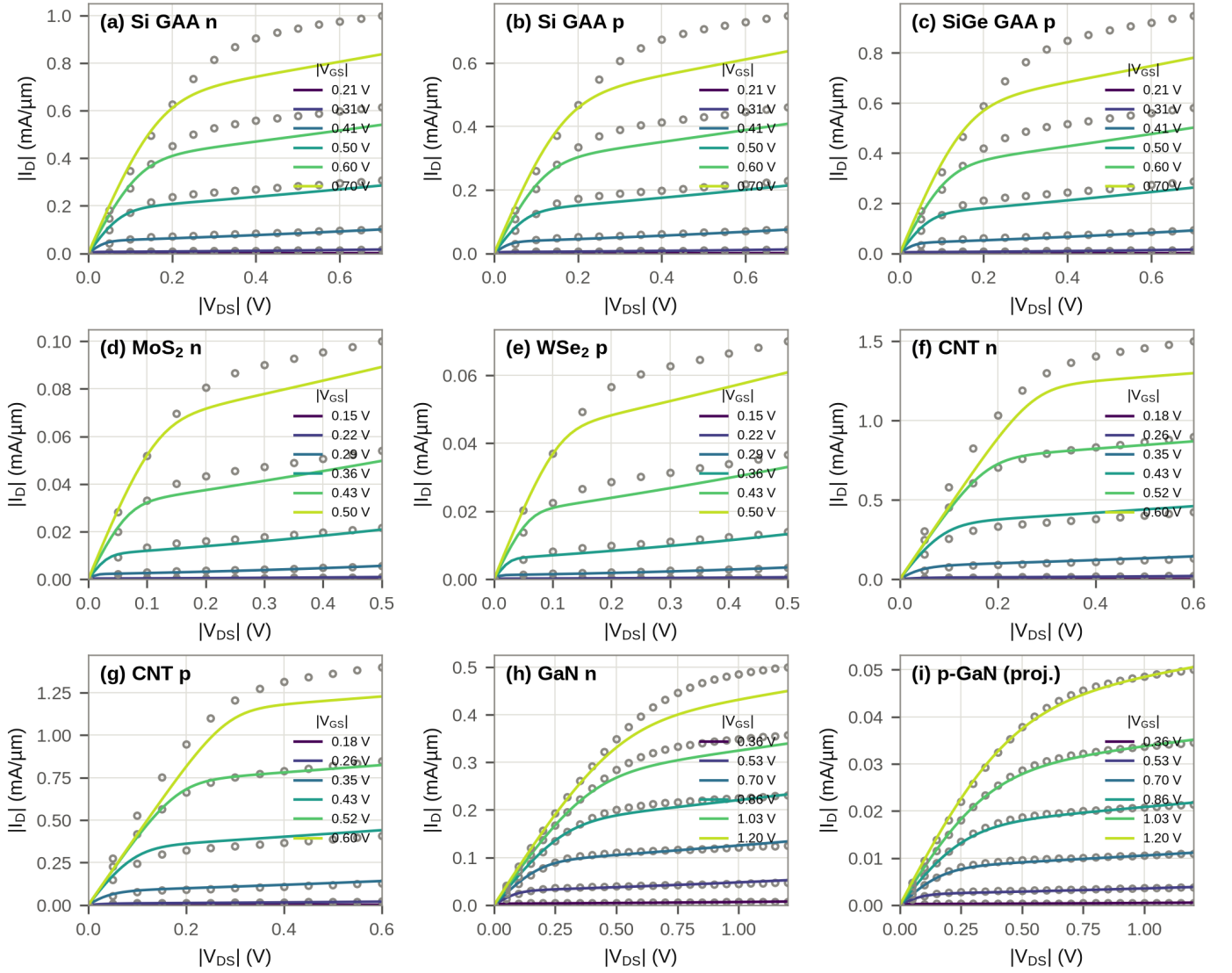


Fig. 3. I_D - V_D families (six gate biases from $0.3V_{DD}$ to V_{DD}) of the calibrated models (lines) and reference (symbols).

TABLE II
CALIBRATED UCM-CFET PARAMETERS (SI VALUES CONVERTED TO THE UNITS SHOWN).

Parameter	Si GAA n	Si GAA p	SiGe GAA p	MoS ₂ n	WSe ₂ p	CNT n	CNT p	GaN n	p-GaN (proj.)
L_g (nm)	16	16	16	20	20	20	20	100	100
V_{TH0} (mV)	346	350	362	343	380	377	375	357	355
n_0	1.14	1.17	1.20	1.18	1.24	1.16	1.15	1.39	1.43
η (mV/V)	46	51	57	70	77	46	46	33	37
μ_0 (cm ² /Vs)	75	55	76	56	63	—	—	300	19
v_{sat} (10 ⁷ cm/s)	2.50	2.00	2.25	1.50	1.20	—	—	3.75	1.25
λ (V ⁻¹)	0.30	0.30	0.30	0.30	0.30	0.00	0.00	0.19	0.08
$R_S W$ ($\Omega \cdot \mu\text{m}$)	53	65	55	500	800	50	60	374	1213
C_{ox} (fF/ μm^2)	43.0	43.0	43.0	34.5	34.5	20.0	20.0	8.0	8.0
C_{gs0} (fF/ μm)	0.30	0.30	0.32	0.20	0.20	0.15	0.15	0.50	0.50
W_{eff} (nm)	150	150	150	100	100	100	100	1000	3000
material block	—	—	—	$C_q=1.2, D_{it}=5e+11$	$C_q=1.1, D_{it}=1e+12$	$T=0.22, \alpha=0.86$	$T=0.21, \alpha=0.86$	$R_{th}=1.5e5$	$R_{th}=1.0e5$

switching energy $\int V_{DD} I_{DD} dt$ over one period) and the N -stage ROs (Fig. 5b).

B. Ring-oscillator measurement

Oscillation is started identically for every platform by an initial condition $V(n_1) = 0$ together with a stated 1% width increase of the stage-1 p-FET. A two-pass transient is used:

a coarse pass, sized from the inverter delay ($T \approx 2Nt_{pd}$), pins down the period T ; the fine pass then runs with a maximum step of $T/300$ and the frequency is measured from the rising $V_{DD}/2$ crossings of $V(n_3)$ between cycles 10 and 30 (the cycle 10–11 value is also recorded and differs by $< 0.01\%$). For GaN the measurement window starts only after five thermal time constants, so that the steady-state self-heating

TABLE III
CALIBRATION ERROR AND FIGURES OF MERIT (MODEL VS. REFERENCE).

Device	MAPE _{lin} (%)	MAPE _{sat} (%)	MAPE _{I_D-V_D} (%)	I _{ON} (mA/μm)	I _{ON} ^{ref} (mA/μm)
Si GAA n	8.1	6.0	10.8	0.84	1.00
Si GAA p	7.9	5.8	10.7	0.64	0.75
SiGe GAA p	8.7	7.0	11.9	0.78	0.95
MoS ₂ n	6.7	5.1	11.2	0.09	0.10
WSe ₂ p	6.9	6.1	12.6	0.06	0.07
CNT n	12.1	10.2	9.9	1.30	1.50
CNT p	11.6	9.7	9.4	1.23	1.40
GaN n	5.1	5.2	5.7	0.45	0.50
p-GaN (proj.)	3.0	1.3	3.7	0.05	0.05

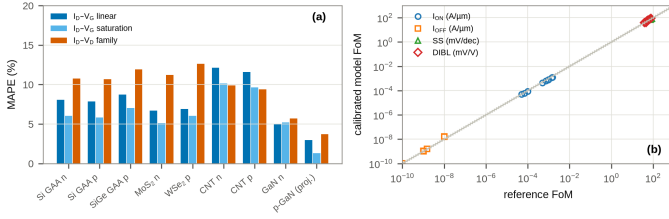


Fig. 4. (a) MAPE per device and data set; (b) parity of model vs. reference figures of merit.

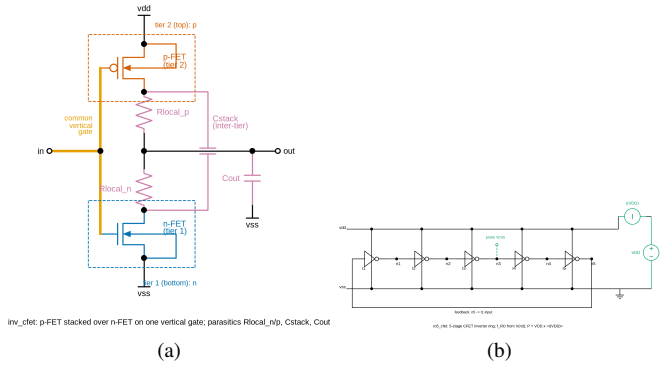


Fig. 5. (a) CFET inverter cell with the explicit inter-tier coupling capacitance C_{stack} , output parasitic C_{out} and local-interconnect resistances $R_{local,n/p}$; (b) 5-stage ring oscillator testbench with the $V(n_3)$ probe and the $I(V_{DD})$ meter.

is included. From f_{RO} the stage delay $t_{pd} = 1/(2Nf_{RO})$, the average power $P_{avg} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} V_{DD} I_{DD} dt$ over the same window, $E_{cycle} = P_{avg}/f_{RO}$ and $PDP = P_{avg}t_{pd}$ follow. Identical window definitions are used for all materials. The ADE Explorer/Assembler equivalents (OCEAN scripts with cross, delay, integ and average calculator functions, parametric sweeps, corners and Monte Carlo sampling) are provided in the companion cadence/ directory.

V. RESULTS

A. Nominal inverters and ring oscillators

Table IV and Figs. 6–8 give the nominal results. All five inverters are functional with V_M/V_{DD} between 0.42 and 0.49, gains between 12 and 22 and both noise margins above $0.28V_{DD}$; the SiGe p-FET moves V_M of the SiGe cell closer to $V_{DD}/2$ than the Si cell (0.49 vs. 0.49) and shortens the low-to-high transition. The 5-stage RO frequencies are 19.39, 21.39, 2.59, 45.39 and 1.69 GHz for Si, SiGe, TMD, CNT and GaN. Stage delays extracted from the 5-, 7- and 9-stage ROs

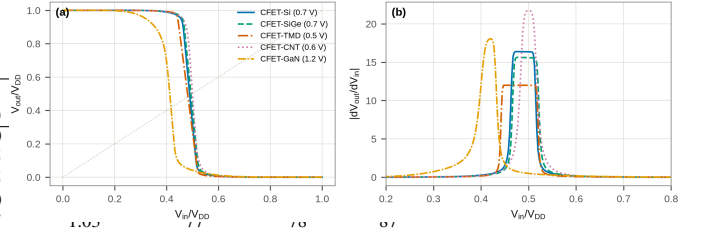


Fig. 6. (a) Normalised voltage transfer characteristics of the five CFET inverters at their nominal V_{DD} ; (b) small-signal gain.

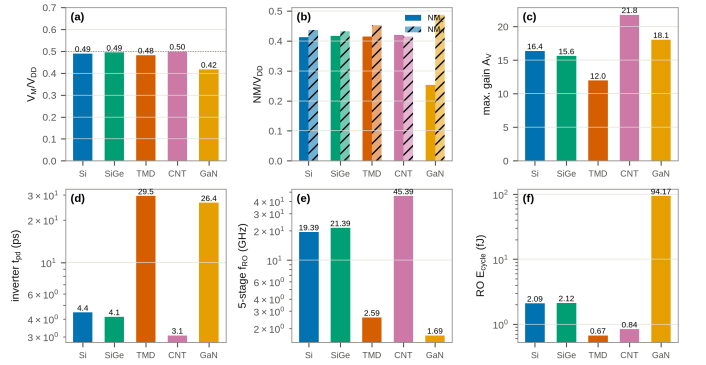


Fig. 7. Nominal figures of merit: (a) V_M/V_{DD} , (b) noise margins, (c) maximum gain, (d) inverter delay, (e) 5-stage RO frequency, (f) energy per RO cycle.

TABLE IV
NOMINAL INVERTER AND 5-STAGE RO FIGURES OF MERIT (27°C).

Platform	V_{DD} (V)	V_M/V_{DD}	A_V	NM_L/V_{DD}	NM_H/V_{DD}	t_{pd}^{inv} (ps)	f_{RO}^{5stg} (GHz)
CFET-Si	0.7	0.490	16.4	0.41	0.44	4.45	19.39
CFET-SiGe	0.7	0.495	15.6	0.42	0.43	4.14	21.39
CFET-TMD	0.5	0.482	12.0	0.41	0.45	29.51	2.59
CFET-CNT	0.6	0.500	21.8	0.42	0.41	3.06	45.39
CFET-GaN	1.2	0.418	18.1	0.25	0.49	26.43	1.69

agree within 1%, which confirms the $1/(2Nf_{RO})$ extraction, while E_{cycle} scales with N as expected. The CNT platform is the fastest (quasi-ballistic injection, small gate charge) and has the second-lowest energy per cycle (0.84 fJ); the TMD platform is $7\times$ slower than Si at 0.5 V because of its 0.5–0.8 kΩ · μm contacts but has the lowest energy per cycle (0.67 fJ); the GaN platform is limited by the projected p-GaN device (hole mobility 20 cm²/Vs, $3\times$ up-sized) to 1.69 GHz and 94.17 fJ at 1.2 V. In the nominal GaN RO the average dissipation per device is only tens of microwatts, so the steady-state self-heating is a few kelvin and changes f_{RO} by $< 0.1\%$; it becomes significant only in the R_{th} and temperature study below.

B. Unified V_{DD} and temperature sweeps

Fig. 9 compares the platforms on common axes. Fig. 10 shows the temperature behaviour.

C. Material-specific sensitivity

- 1) Si and SiGe:
- 2) MoS₂/WSe₂:
- 3) CNT:

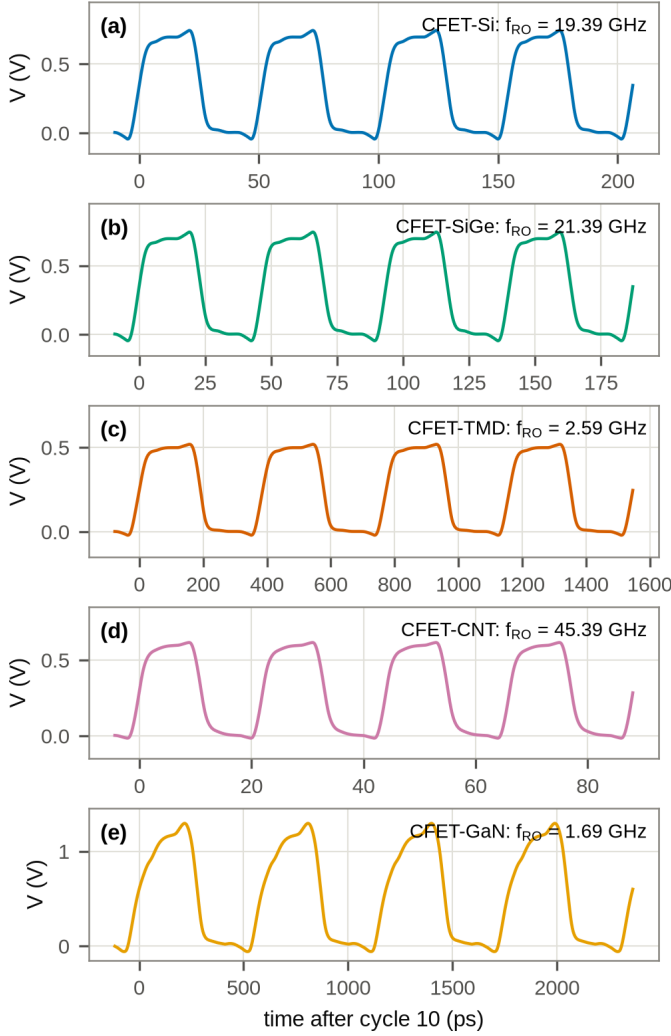


Fig. 8. 5-stage RO waveforms $V(n_3)$ for the five platforms, four cycles after cycle 10 (the measurement window).

TABLE V
MONTE CARLO SUMMARY AND YIELD (CRITERIA: $f_{RO} \geq 0.9f_{RO,nom}$;
 $N_{ML,H} > 0.1V_{DD}$ AND $A_V > 10$).

pending

4) GaN:

D. Monte Carlo and yield

Monte Carlo was run only after the nominal VTCs, RO start-up and extraction had been verified for all platforms. The random variables are the physical model parameters listed in Section III (Si: V_{TH0} with $\sigma = 20$ mV including work-function shift, W_{NS} with $\sigma = 1$ nm, R_{SD} 10%; SiGe adds 10% on the p-mobility as a composition proxy; TMD: R_C 30% log-normal, V_{TH0} 50 mV, mobility 20%, D_{it} 0.3 decades; CNT: density 15%, metallic fraction log-normal around 10^{-6} with 0.5 decades, R_C 30%; GaN: V_{TH0} 50 mV including trap shift, mobility 10%, R_{th} 20%), drawn independently for the n- and p-device, with the same seeded generator for every platform.

TABLE VI
SELECTION MAP DERIVED FROM THE SIMULATIONS.

Design need	Preferred CFET family
Highest maturity / main logic baseline	CFET-Si
Better p/n balance with Si-compatible processing	CFET-SiGe
Lowest V_{DD} , leakage-sensitive logic	CFET-TMD, if R_C is below the crossover value (Fig. 12)
Highest performance post-Si research	CFET-CNT, if purity and density are controlled (Fig. 13)
High-temperature / integrated power-control logic	CFET-GaN (projected p-GaN)

E. Conceptual layout

Fig. 16 shows conceptual layouts generated with a generic 3-nm-class CFET rule set (contacted gate pitch 45 nm, metal pitch 24 nm, 20-nm-wide nanosheets, four-track cell height because the p-tier folds over the n-tier). The bottom n-sheets, top p-sheets, common gate, tier contacts, inter-tier via joining the two drains and the M0 power rails are separate layers; a boolean DRC (minimum width and spacing per layer, gate-to-contact spacing, via enclosure) reports zero violations for the five inverters, the five 5-stage ROs and the top cell. The Si/SiGe inverter occupies $90 \times 96 \text{ nm}^2$; the TMD/CNT projections $108 \times 96 \text{ nm}^2$; the GaN projection $400 \times 400 \text{ nm}^2$. These layouts annotate the gate length, sheet pitch, tier spacing and the location of C_{stack} ; they are not tape-out-ready, which would require a real CFET PDK with DRC/LVS decks.

VI. MATERIAL-SELECTION MAP

Fig. ?? condenses the results.

VII. LIMITATIONS

The reference data are literature-anchored reconstructions, not measurements; the MAPE values quantify the model's ability to follow an independent formulation, not agreement with hardware, and must be re-established with measured or TCAD curves before the models are used for a specific technology. The p-GaN device is a projection. The TMD block omits Schottky-barrier bias dependence of the contacts and the CNT block uses a single transmission parameter; the charge model uses a symmetric end-point partition rather than a full Ward-Dutton partition; CFET parasitics are lumped. The sensitivity and Monte Carlo trends, which are the main results, are driven by the physical parameters of the model and are therefore expected to be robust to these simplifications, but the absolute numbers should be read as platform-level projections.

VIII. CONCLUSION

A single compact-model core with four material blocks, implemented once in Verilog-A for Spectre/ADE and mirrored in ngspice, allows Si, SiGe, MoS₂/WSe₂, CNT and GaN CFET logic to be calibrated, simulated and compared with one flow and one set of measurement definitions. The flow reproduces a 5-stage RO at 19.39 GHz and 2.09 fJ/cycle for the Si baseline and yields a material-selection map: SiGe for p/n balance, TMD for the lowest energy at low V_{DD} provided the contact

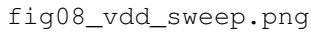


Fig. 9. Unified V_{DD} sweeps of the 5-stage ROs: (a) f_{RO} vs. V_{DD} , (b) power vs. frequency, (c) energy per cycle vs. frequency, (d) PDP vs. V_{DD} .

resistance stays below the crossover value, CNT for the highest speed provided the metallic-tube fraction stays below 10^{-3} , and GaN for high-temperature operation at an energy penalty set by the p-channel device. All models, scripts, layouts and results are released for reuse.

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fig09_temperature.png

Fig. 10. Temperature sweeps at nominal V_{DD} (-40 to 125°C , GaN to 200°C): (a) normalised f_{RO} , (b) energy per cycle, (c) inverter static power.

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fig10_si_sige_sensitivity.png

Fig. 11. Si/SiGe study: 5-stage f_{RO} and V_M/V_{DD} vs. the number of nanosheets in the n- and p-tier for (a) CFET-Si and (b) CFET-SiGe; (c) effect of the SiGe/Si hole-mobility ratio on the rise/fall delay mismatch, V_M and f_{RO} .

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fig11_tmd_contact_crossover.png

Fig. 12. TMD study: (a) iso-frequency energy-per-cycle ratio E_{TMD}/E_{Si} vs. the n- and p-contact resistance (the Si energy is interpolated at the TMD frequency from the Si V_{DD} sweep extended to 0.3 V); (b) f_{RO} vs. $R_{C,n}$; (c) effect of the interface trap density.

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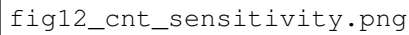


fig12_cnt_sensitivity.png

Fig. 13. CNT study: normalised f_{RO} and PDP (left axes) and inverter static power (right axes) vs. (a) tube density, (b) metallic-tube fraction, (c) contact resistance.

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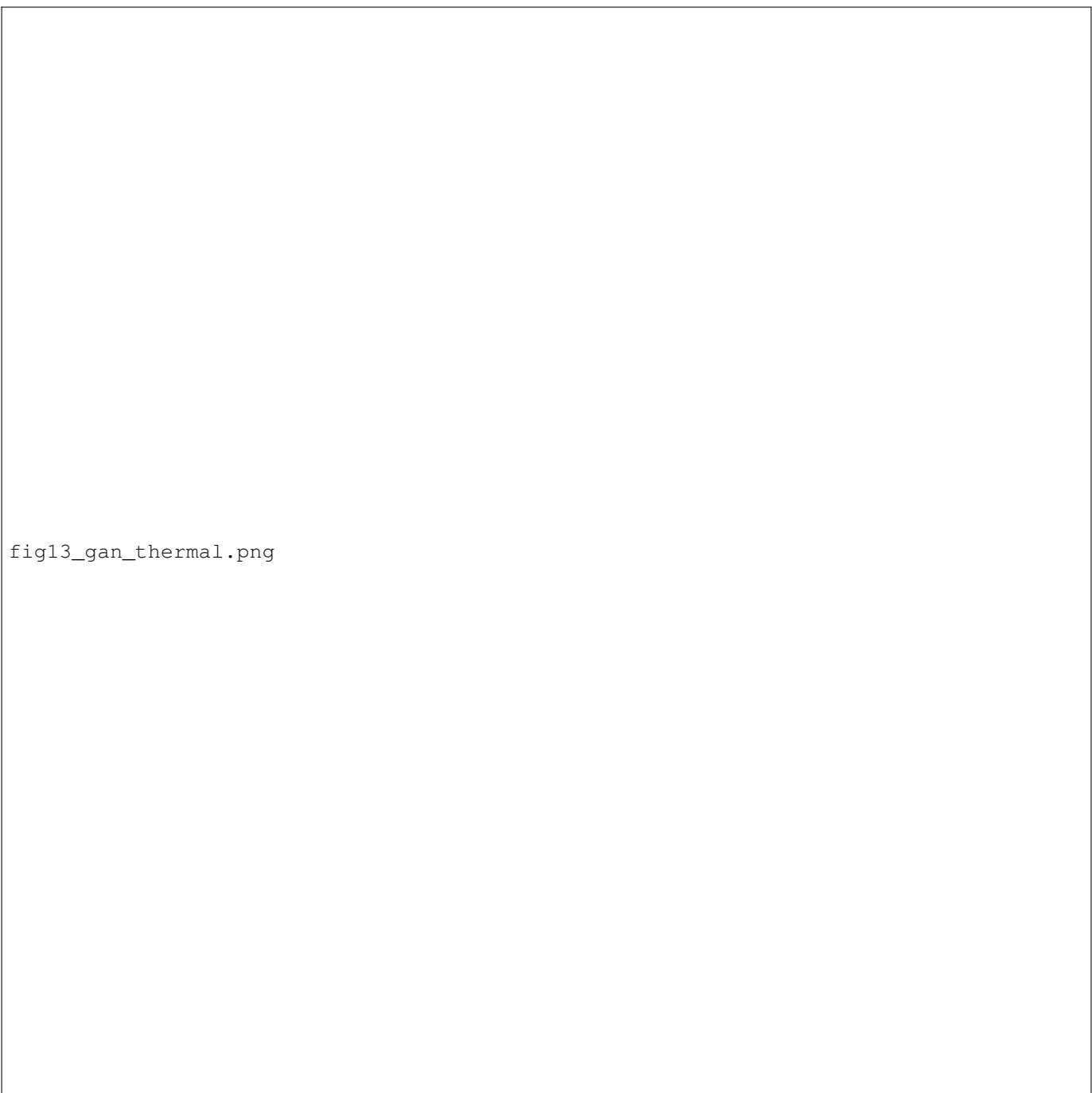


fig13_gan_thermal.png

Fig. 14. GaN study: (a) f_{RO} and static power of CFET-GaN and CFET-Si from 300 to 500 K; (b) self-heating delay penalty vs. thermal resistance at three temperatures; (c) f_{RO} vs. common and skewed threshold-voltage shifts.

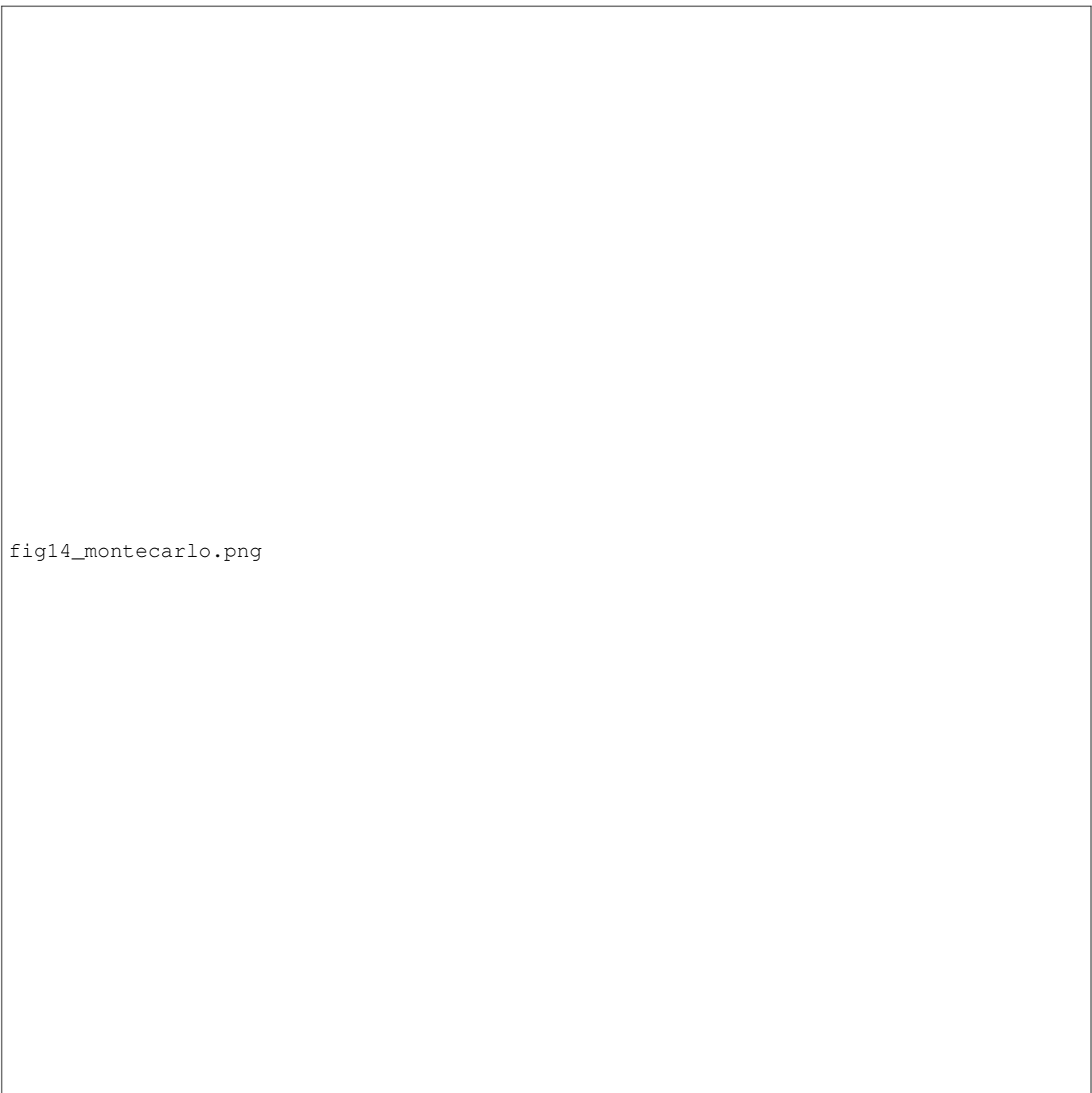


fig14_montecarlo.png

Fig. 15. Monte Carlo (200 RO samples and 1000 inverter samples per platform): (a) f_{RO} normalised to nominal, (b) energy per cycle normalised to its median, (c) V_M/V_{DD} , (d) noise-margin scatter with the $0.1V_{DD}$ functional limits, (e) frequency yield Y_f and functional yield, (f) relative frequency spread.

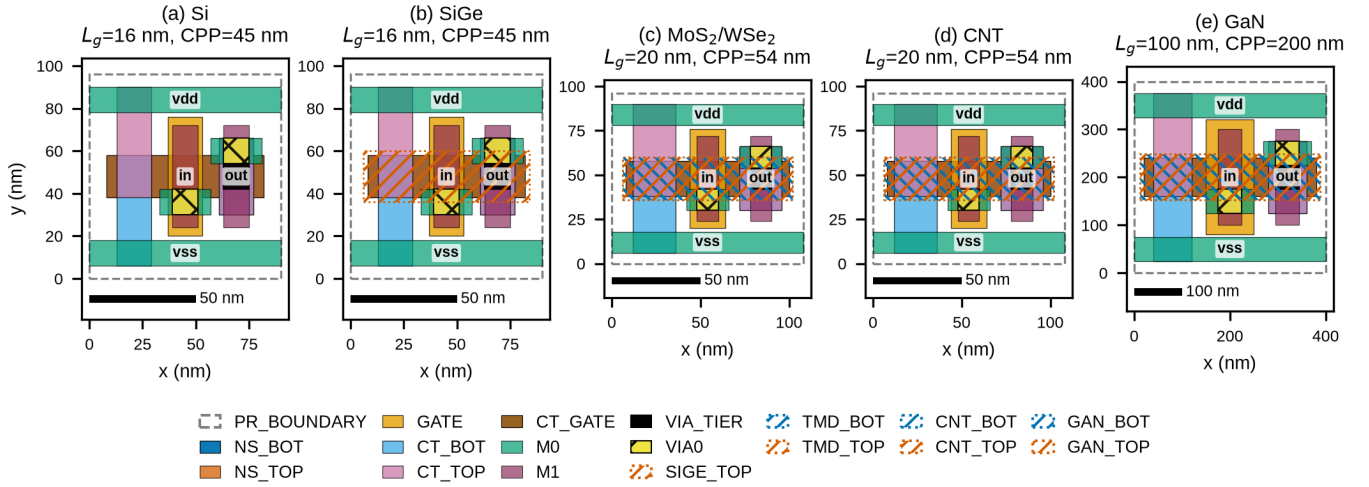


Fig. 16. Conceptual, DRC-clean GDSII layouts of the five CFET inverter cells (plan view; tiers drawn as separate layers): Si and SiGe at $L_g = 16$ nm, $CPP = 45$ nm, 90×96 nm cell; TMD and CNT projected at $L_g = 20$ nm, $CPP = 54$ nm; GaN projected at $L_g = 100$ nm, $CPP = 200$ nm.

